

Development of an Enterprise Infrastructure Management System (EIMS)

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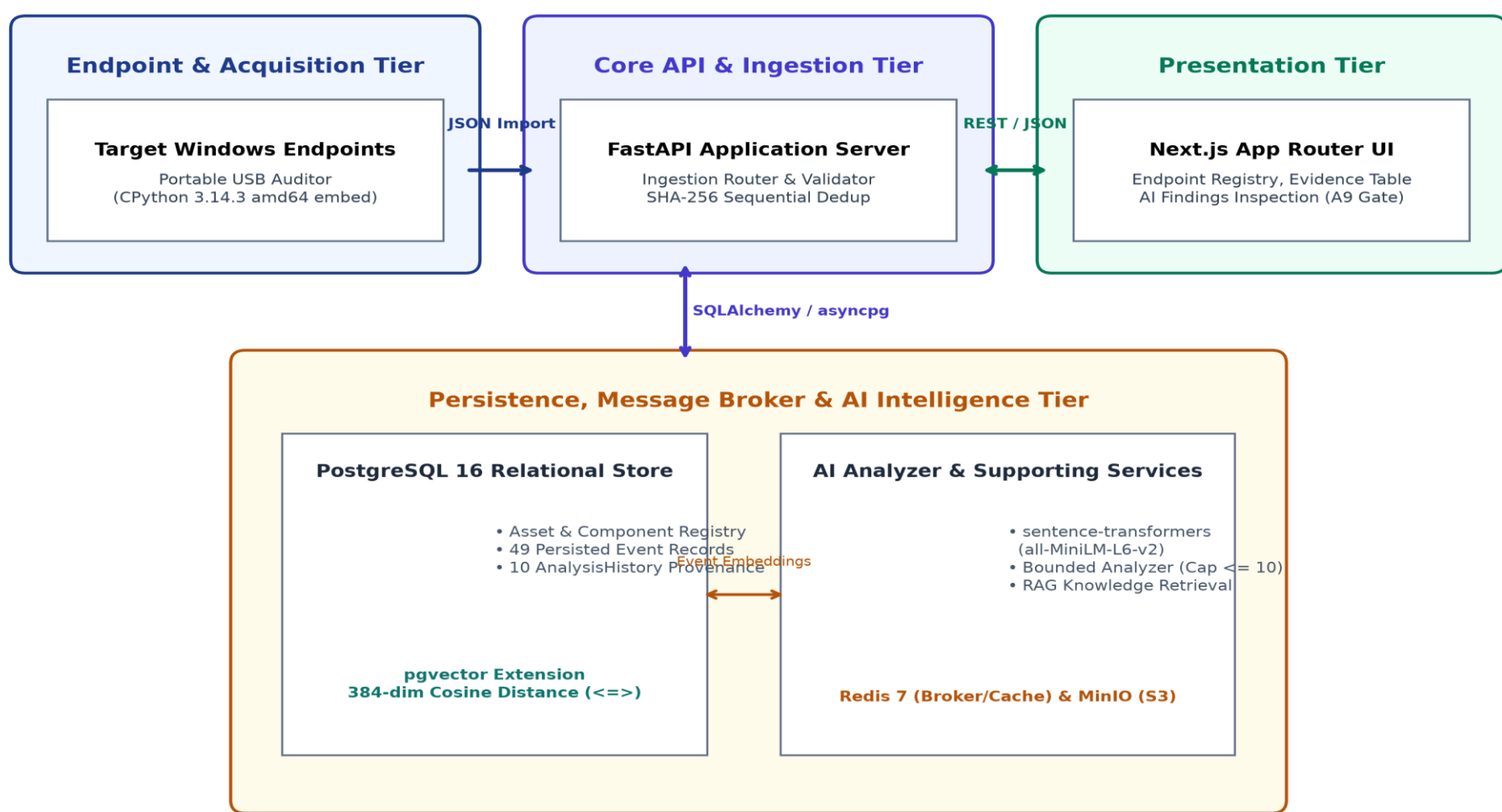
1. Background & Problem Statement

- Network Enclave Isolation:** High-security enterprise servers prohibit continuous outbound monitoring connections.
- Fragile Ephemeral Identity:** DHCP leases and host renames break asset history without hardware-anchored identity.
- Manual Diagnostic Friction:** Ad-hoc scripts produce fragmented, unverified diagnostic notes without provenance.
- Telemetry Volume Overload:** Windows Event Logs generate thousands of routine records daily, slowing fault triage.

2. Project Engineering Objectives (Chapter 1)

- OBJ-1 Central Asset Registry:** Hardware composite fingerprints anchoring canonical server identity.
- OBJ-2 Portable USB Auditor:** Embedded CPython 3.14.3 runtime executing non-destructive read-only queries.
- OBJ-3 Offline Ingestion Pipeline:** Schema-validated normalization of offline JSON evidence reports.
- OBJ-4 Deterministic Deduplication:** Application-level sequential SHA-256 deduplication and severity queue.
- OBJ-5 Bounded AI Investigation:** pgvector cosine similarity retrieval for prioritized events (cap <= 10).
- OBJ-6 Web Dashboard:** Responsive Next.js 14 App Router UI for asset telemetry, evidence, and AI findings.
- OBJ-7 Empirical Validation:** End-to-end multi-tier verification from USB extraction to live browser display.

3. Multi-Tier System Architecture (Hybrid Modular Monolith)



- Hybrid Modular Monolith:** Consolidates collection, ingestion, dedup, and AI triage in a single maintainable service without microservice networking overhead.
- Persistence & Vector Separation:** PostgreSQL 16 handles ACID relational entities while pgvector computes 384-dimensional cosine distance in unified SQL queries.
- Decoupled Supporting Tier:** Redis manages message queuing/caching, MinIO provides S3-compatible storage, and Next.js 14 delivers hybrid server/client rendering.

4. Core Technology Implementation Stack

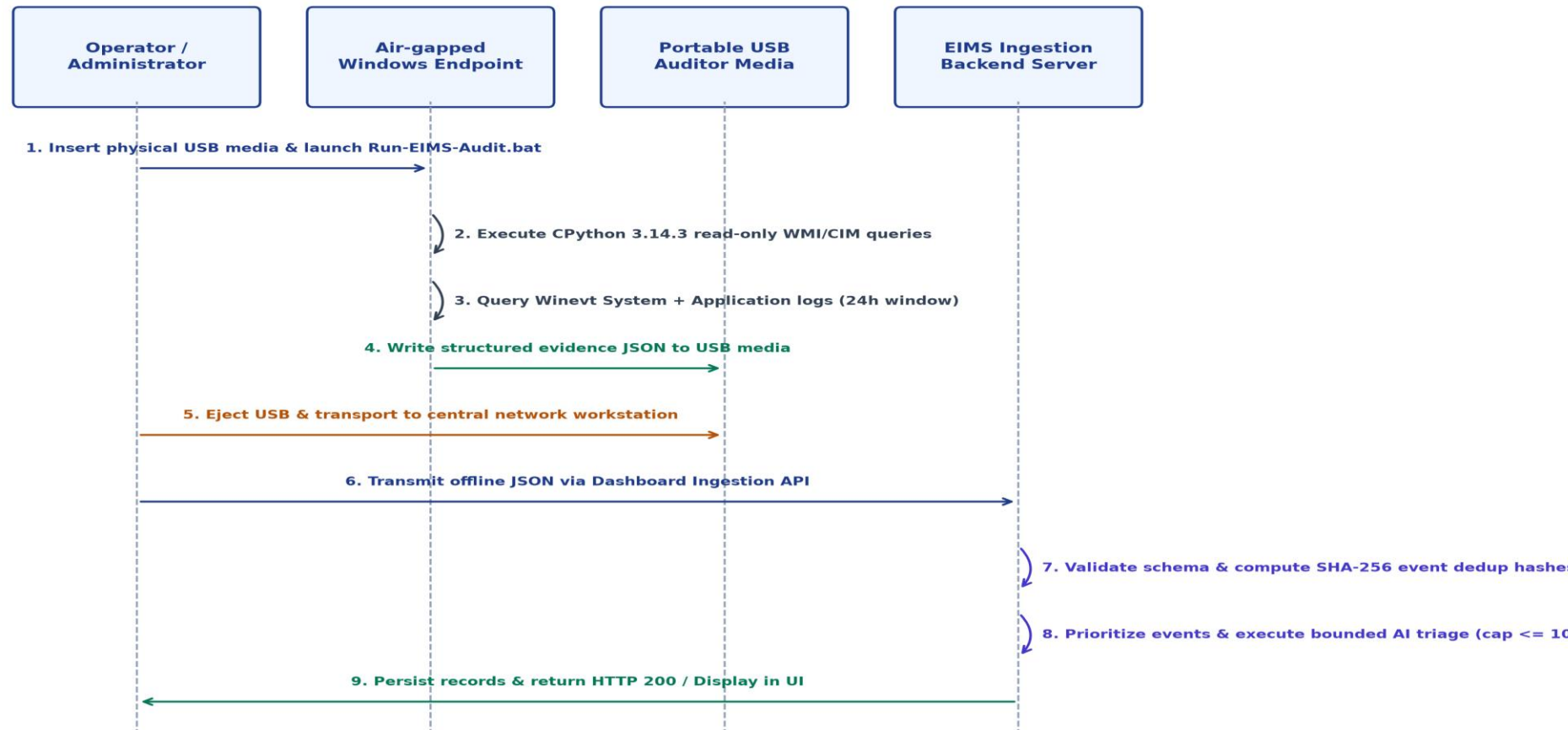
- Backend API Gateway:** FastAPI + Pydantic + Starlette delivering REST APIs and schema validation.
- Database & Vector Store:** PostgreSQL 16 with pgvector (384-d dense vector cosine distance search).
- Broker & Object Store:** Redis 7 (caching/queues) and MinIO (S3-compatible storage for OCR/specs).
- Operator Web Dashboard:** Next.js 14 App Router, React 18, TypeScript, and Tailwind CSS.
- Portable USB Auditor:** Embedded CPython 3.14.3 amd64 distribution (no target dependencies).
- AI Embedding Hierarchy:** FastEmbed BAAI/bge-small-en-v1.5 primary / SentenceTransformer fallback.

5. Event Deduplication & Bounded AI Role

DedupKey = SHA-256(asset_id || channel || provider || record_id || occurrence_time)

- Sequential Dedup Integrity:** Windows record_id is monotonically assigned; hash enables deterministic duplicate skipping across sequential imports.
- Concurrency Boundary:** Enforced sequentially in memory. Concurrent parallel imports may race as database-level uniqueness is not enforced.
- Severity Ranking:** Candidate events prioritized: Critical (Level 1) > Error (Level 2) > Warning (Level 3) > Info (Level 4).
- Bounded Workload Cap:** Analyzer strictly triages at most 10 selected new events per import, protecting worker threads.
- Assistive Role:** AI provides diagnostic hypotheses and suggested verification checks; does NOT execute autonomous remediation.

6. Portable Disconnected Collection Workflow



- Bundled Portable Runtime:** Embedded CPython 3.14.3 amd64; requires NO preinstalled Python, pip, Git, or Docker on target hosts.
- Non-Destructive Posture:** Read-only WMI/CIM and event queries; writes reports/logs strictly to configured USB directories.
- Scoped Telemetry Window:** Captures System and Application logs over 24h (default maximum 500 events across combined query).
- Deterministic JSON Output:** Produces structured, schema-compliant JSON payloads for central offline import.

7. Empirical Validation Results & Measured Metrics

37 / 0 / 4Hermetic Test Suite
37 passed, 0 failed, 4 warnings**49 Events**Persisted Field Telemetry
34 System + 15 Application**10 Records**AnalysisHistory Cap
10/10 prioritized events triaged**HTTP 200**Filtered History API
GET /api/v1/history?asset_id=...**Gate A9 PASS**Browser Runtime Gate
49 events + active AI findings**0 Errors**Frontend Quality Gates
Strict tsc & Next.js build PASS

- Workload Bounding Distinction:** The 49 events represent persisted evidence in the database. The analyzer is strictly bounded to at most 10 selected new events per import (not claiming all 49 were analyzed by AI).
- Validation Scope Boundary:** Earlier field trials verified physical USB execution and JSON extraction. The latest on-site session verified downstream data integrity (persisted DB rows, filtered API, and UI) without re-running physical USB collection.
- Referential Provenance:** Every AnalysisHistory recommendation maintains explicit referential linkage back to originating event_id, asset_id, and collection timestamp.

8. Scoped Limitations & Operational Boundaries

- Windows-First Focus:** Portable collection targets Windows WMI/winevt subsystems; Linux/macOS collectors deferred.
- Scoped Telemetry Channels:** Constrained to System and Application logs; high-volume Security authentication logs excluded.
- Sequential Dedup Only:** Application-level sequential deduplication; does not provide distributed concurrent database locks.
- Bounded Triage Capacity:** Analyzer cap strictly limits AI triage to at most 10 prioritized events per batch import.
- Assistive Role:** AI suggests diagnostic hypotheses; does not act as autonomous authority or execute automated remediation.
- Academic Prototype Baseline:** Evaluated single-instance prototype baseline; not a commercial SIEM or certified forensics suite.

9. Project Conclusion & Future Roadmap

Summary of Key Achievements:

- Centralized Infrastructure Governance:** Unified asset tracking anchored by cryptographic hardware composite fingerprints.
- Air-Gapped Evidence Acquisition:** Non-destructive offline collection utilizing self-contained embedded CPython 3.14.3.
- Deterministic Deduplication & Bounded Triage:** Enforced sequential SHA-256 event fingerprinting and 10-event RAG triage.
- Empirically Validated Pipeline:** Confirmed end-to-end data integrity across 49 field events, HTTP 200 API, and Gate A9.

Future Engineering Roadmap:

- Database-Level Concurrency:** Enforce PostgreSQL composite UNIQUE constraints with ON CONFLICT DO NOTHING for parallel imports.
- Cross-Endpoint Event Correlation:** Implement multi-host temporal clustering to detect infrastructure-wide incident chains.
- Enterprise Directory Federation:** Integrate SAML 2.0 / OIDC single sign-on with corporate Active Directory domain controllers.
- Cross-Platform Collection:** Develop lightweight portable collectors for Linux systemd journals and macOS logging subsystems.